

Exercise1 : Dealing with one disturbance

Let us consider a system described by the following block diagram :

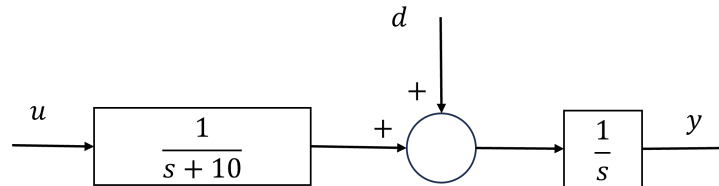


FIGURE 1 – Block Diagram of the studied system

Part 1 :State-feedback with disturbance cancellation

The specifications are :

- overshoot : 5%
- time response : 2 seconds
- No steady-state error

First, let us suppose that the state x and the disturbance d is measured.

1.1 Determine a state-feedback control of the following form that ensures the specifications

$$u = -Kx + ly^r + Nd$$

Now, let us suppose that only the output y is measurable. We will also assume that the disturbance is constant.

- 1.2** Define an augmented state-space to integrate the disturbance as an additional state-variable.
- 1.3** Check whether this augmented system observable? Propose to design a state-observer to estimate both x and d
- 1.4** Using the joint Matlab-Simulink files, implement this solution.

Part 2 : State-feedback with integral action

We now want to implement another control law that will cancel the steady-state error. To do this, we propose to determine a state-feedback control with an integral action :

$$u = -Kx + k_i \int (y^r - y)$$

- 2.1** Check if the conditions of existence are satisfied
- 2.2** Implement the solution on Matlab/Simulink using the joint files, and discuss about the influence of the parameter k_i .
- 2.3** Investigate on the poles of the closed-loop system for different values of k_i .

Part 3 : State-feedback with integral action : simultaneous determination of K and k_i

We still want to implement a state-feedback control with an integral action, but not in a two-step procedure AND using a LQ approach.

$$u = -Kx + k_i \int (y^r - y)$$

- 3.1** Think on a way to determine at the same time K and k_i using LQ control.
- 3.2** Implement the solution on Matlab/Simulink using the joint files, and discuss about the influence of the parameters Q and R of the criterion.

Part 4 : Convergence analysis

- 4.1** Determine the final value of the output for all the different strategies and check whether the theory is consistent with what you have observed in your simulations.